An Internship Report

Of

DATA ANALYSIS

Submitted

To

IILM UNIVERSITY

In partial fulfilment of the requirement of degree of

B.Tech (CSE)



By

Roll Number of Student: 2410031256

Name of Student: ARYAN SINGH

Batch: 2024–28

CANDIDATE'S DECLARATION

I, ARYAN SINGH, do hereby declare that the internship report titled “Data Analysis Internship at ElevenMonk” has been prepared as part of the requirements of the Bachelor of Technology in Computer Science and Engineering at IILM University. The report presents the work and learning activities completed during my internship in the Data Analysis domain from 3 August 2026 to 29 August 2026.

The project descriptions in this report are based on the internship tasks carried out during the internship at ElevenMonk. The attached completion certificate is included as a supporting document.

Signature: _____

Student Name: ARYAN SINGH

Roll No.: 2410031256

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to ElevenMonk for providing me with the opportunity to undertake an internship in the field of Data Analysis. The internship gave me a practical environment in which I could apply data analysis concepts to the kind of data generated by an IT solutions company.

I am thankful to the founder, Ms. Bina Singh, and the team at ElevenMonk for their guidance and support throughout the internship, and for exposing me to how website development and app development projects generate data that can be analysed to support better decisions.

I also express my gratitude to IILM University and the faculty of the Computer Science and Engineering programme for their academic support and encouragement.

Finally, I am thankful to my parents and everyone who supported me throughout the internship and helped me complete the assigned projects successfully.

Signature: _____

Student Name: ARYAN SINGH

Roll No.: 2410031256

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3. INTERNSHIP COMPLETION CERTIFICATE

The completion certificate issued by ElevenMonk is reproduced below.

ELEVENMONK

CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that

Aryan Singh

has successfully completed an internship as **Data Analyst Intern** at ElevenMonk, from **3rd August 2026** to **29th August 2026**.

During this period, he demonstrated dedication, professionalism, and a strong willingness to learn, and contributed meaningfully to the team's work.

We wish him continued success in all future endeavors.

29th August 2026

Date

Bina Singh

Bina Singh

Founder, ElevenMonk

ElevenMonk | Varanasi, India | www.elevenmonk.com | info@elevenmonk.com

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents a detailed account of the Data Analysis internship undertaken at ElevenMonk from 3 August 2026 to 29 August 2026. According to the internship completion certificate, the role was Data Analyst Intern. ElevenMonk is an IT solutions company based in Varanasi, India, and its work gives a natural setting for data analysis, because every website and application it delivers produces data about users, traffic, performance and projects.

The internship was organized around four projects completed over four weeks. Together, these projects covered the major stages of a data analysis workflow: collecting and cleaning data, exploring it visually and statistically, querying it with SQL and reporting key metrics, and finally combining everything in an end-to-end business analytics case study.

4.2 Organization Profile

ElevenMonk is an IT solutions company that provides services such as website building and app development, along with further IT services for its clients. The company is located in Varanasi, India, and can be reached through www.elevenmonk.com and info@elevenmonk.com. The internship completion certificate was issued by the founder, Ms. Bina Singh.

Organizations of this kind work on several client projects at once. Each project creates data such as website visits, user behaviour inside apps, enquiries and leads, project timelines and support requests. A data analyst helps turn this data into clear information that supports planning and decision-making.

4.3 Problem Statement

Data from real business activity is rarely ready for immediate analysis. It can contain missing values, duplicate records, outliers, inconsistent formats and variables that need careful interpretation. For an IT solutions company, data about website traffic, app usage and client projects is often spread across different sources and is not organized for decision-making.

At the same time, useful analysis requires more than cleaning data: patterns must be explored, relationships visualized, key metrics tracked and conclusions communicated clearly. The internship addressed this practical problem through a sequence of projects covering data preparation, exploration, reporting and a complete case study.

4.4 Project Objectives

- To practice data cleaning by handling missing values, outliers, duplicate records and inconsistent formats.
- To use Python libraries such as Pandas, NumPy, Matplotlib and Seaborn for analysis and visualization.
- To query and summarize structured data using SQL.
- To explore datasets statistically and visually to identify patterns, trends and correlations.
- To define and report key performance indicators (KPIs) through dashboards and summary reports.
- To present analytical findings and recommendations in a structured, understandable form.
- To apply the complete analysis workflow to a business case relevant to web and app services.

4.5 Scope of the Internship

The scope covered practical data analysis activities from raw-data preparation to reporting and business recommendations. It included data cleaning, exploratory data analysis, visualization, SQL-based analysis, KPI reporting and an end-to-end case study.

This report describes the work at the level of tasks, methods and learning. It does not disclose confidential client data and does not claim specific dataset names, numerical business results or production-system deployment.

4.6 Technologies and Tools Used

Technology / Tool	Purpose in the Internship
Python	Primary programming language for data processing, analysis and visualization.
Pandas	Data loading, cleaning, transformation and tabular analysis.
NumPy	Numerical computation and statistical calculations.
Matplotlib	Creation of charts and analytical plots.
Seaborn	Statistical and exploratory visualizations.
SQL	Querying, filtering, joining and aggregating structured data.
Microsoft Excel	Quick data checks, pivot tables and summary reports.
Jupyter Notebook	Interactive environment for analysis and documentation.

4.7 Overall Data Analysis Workflow

The four projects can be viewed as a connected workflow. The first project focuses on collecting and preparing data. The second explores the data and visualizes patterns. The third turns data into KPIs and reports using SQL. The fourth applies the whole approach to a business case.

1. Raw / Given Dataset
2. Data Collection & Cleaning
3. Exploratory Analysis
4. Visualization & Pattern Identification
5. SQL Analysis & KPI Reporting
6. Interpretation of Results
7. Insights, Recommendations & Structured Presentation

4.8 Project 1: Data Collection, Cleaning & Preparation

Due date: 07 August 2026 | Status: Completed Successfully

The first project focused on working with raw, business-style data of the kind produced by website and app projects — for example traffic records, user activity logs and enquiry or lead-tracking sheets — and preparing it for analysis.

Objectives

- Collect and combine raw data from different file sources.
- Identify and handle missing values.
- Detect and manage outliers and duplicate records.
- Standardize formats such as dates, categories and text fields.
- Use Pandas and NumPy for cleaning and transformation.

Approach

The task begins with inspection of the dataset structure and data quality. Cleaning steps are then applied: removing duplicates, treating missing values with suitable methods, checking outliers and correcting inconsistent entries. The cleaned data is validated against the original to confirm that no useful information was lost, and a clean, analysis-ready dataset is saved for the next project.

Learning Outcome

This project develops practical understanding of data preprocessing and the importance of data quality, which is the foundation for every later analysis.

4.9 Project 2: Exploratory Data Analysis & Visualization

Due date: 14 August 2026 | Status: Completed Successfully

The second project concentrated on Exploratory Data Analysis (EDA). Its purpose was to understand the cleaned data through statistical summaries and visualizations before drawing any conclusions.

Objectives

- Generate statistical summaries of the data.
- Study distributions of individual variables.
- Examine relationships and correlations between variables.
- Create clear charts using Matplotlib and Seaborn.
- Identify patterns, trends and influencing factors.

Approach

The analysis moves through dataset understanding, summary statistics, univariate analysis (distributions), bivariate analysis (relationships and correlations) and finally the summary of findings. Histograms, bar charts, box plots, line charts and heatmaps are used to make trends and comparisons easy to interpret.

Learning Outcome

The expected outcome was stronger analytical thinking and the ability to communicate patterns in data visually.

4.10 Project 3: SQL Analysis & KPI Dashboard Reporting

Due date: 21 August 2026 | Status: Completed Successfully

The third project focused on retrieving and summarizing data with SQL and presenting key performance indicators (KPIs) in a dashboard-style report that a non-technical reader can understand.

Objectives

- Write SQL queries using SELECT, WHERE, GROUP BY, ORDER BY and JOIN.
- Aggregate data to produce summary metrics.
- Define relevant KPIs, such as monthly traffic, enquiry or lead counts and project status counts.
- Build dashboard-style visual reports using charts and summary tables.
- Present results in a structured and readable format.

Approach

Data is first organized into related tables. SQL queries are then used to answer specific business questions, and the query results are visualized as KPI summaries and charts. Each KPI is accompanied by a short explanation so the report can be used directly for review and discussion.

Learning Outcome

The expected outcome was the ability to move from raw tables to decision-ready reports using SQL and visualization.

4.11 Project 4: Business Analytics Case Study

Due date: 28 August 2026 | Status: Completed Successfully

The fourth project was an end-to-end business analytics case study relevant to ElevenMonk's field of work: analysing website and app performance data to understand how users engage with digital products and which factors influence results.

Objectives

- Define the business question and the data needed to answer it.
- Clean and prepare the available data.
- Perform exploratory analysis and compute relevant metrics.
- Visualize key findings clearly.
- Draw conclusions and give practical recommendations.

Approach

The case study follows the complete workflow: problem understanding, data collection, data processing, analysis and visualization, and finally results and recommendations. Findings are summarized in a structured report so that decision-makers can understand what the data shows and what actions it suggests.

Learning Outcome

The expected outcome was the ability to apply data analysis skills end-to-end on a realistic business problem.

4.12 Methodology

The internship followed a project-based methodology spread across four weeks, from 3 August to 29 August 2026. Each week was built around one project, with each project building on the previous one.

Project	Focus	Due Date	Status
1	Data Collection, Cleaning & Preparation	07 Aug 2026	Completed Successfully
2	Exploratory Data Analysis & Visualization	14 Aug 2026	Completed Successfully
3	SQL Analysis & KPI Dashboard Reporting	21 Aug 2026	Completed Successfully
4	Business Analytics Case Study	28 Aug 2026	Completed Successfully

Integrated Methodology

- Understand the project requirement and the given data.
- Inspect and clean the data where required.
- Use statistical analysis and visualization to understand the dataset.
- Use SQL to retrieve and summarize data and define KPIs.
- Interpret the results in the context of the business question.
- Present findings and recommendations in a structured manner.

4.13 Expected Outcomes

- Improved understanding of practical data cleaning and preparation.
- Ability to use Pandas, NumPy, Matplotlib and Seaborn for analysis and visualization.
- Ability to query structured data with SQL.
- Improved ability to identify patterns, trends and correlations through EDA.
- Experience in defining KPIs and building clear reports.
- Experience in presenting analytical findings clearly and systematically.
- Ability to apply the data analysis workflow to a real business problem.

Skills Developed

Area	Skills / Exposure
Data Preparation	Missing values, outliers, duplicates, standardization
Data Analysis	Statistical summaries, trends, correlations, KPIs
Visualization	Matplotlib, Seaborn, dashboard-style reporting
Data Querying	SQL selection, filtering, joins and aggregation
Business Insight	Linking data findings to business decisions
Communication	Structured reporting and presentation of insights

Overall, the internship provided a progression from foundational data preparation to analysis, reporting and a complete business case study. All four assigned projects were completed successfully.

4.14 Certificate and Verification Details

The completion certificate issued by ElevenMonk confirms that Aryan Singh successfully completed an internship as Data Analyst Intern from 3rd August 2026 to 29th August 2026. The certificate is dated 29th August 2026 and is signed by Ms. Bina Singh, Founder, ElevenMonk.

The organization’s details as printed on the certificate are: ElevenMonk, Varanasi, India; www.elevenmonk.com; info@elevenmonk.com.

The original completion certificate is included in the earlier pages of this report as documentary proof.

5. BIBLIOGRAPHY / REFERENCES

1. ElevenMonk, “Certificate of Internship Completion,” Aryan Singh, Data Analyst Intern, 29 August 2026.

2. ElevenMonk, Internship project: “Data Collection, Cleaning & Preparation,” due 07 August 2026.

3. ElevenMonk, Internship project: “Exploratory Data Analysis & Visualization,” due 14 August 2026.

4. ElevenMonk, Internship project: “SQL Analysis & KPI Dashboard Reporting,” due 21 August 2026.

5. ElevenMonk, Internship project: “Business Analytics Case Study,” due 28 August 2026.

6. Pandas and NumPy documentation used for general reference.

7. Matplotlib and Seaborn documentation used for general visualization reference.

8. SQL language reference material used for general reference.